How Do You Build a CNOT Gate from Rydberg Atoms?

Literature review note[1]

[1]Seeded by Farouk et al. 2023 · 13 references collected

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Abstract

Neutral atoms trapped in optical tweezers are a leading platform for quantum computing, and two-qubit entangling gates — particularly the CNOT — are the critical bottleneck. This note traces three converging research threads that shape modern Rydberg CNOT gate design: **(i)** adiabatic passage through Stark-tuned Förster resonances as the foundational interaction mechanism, **(ii)** two-photon laser excitation pathways that make Rydberg states experimentally accessible, and **(iii)** counter-diabatic driving techniques that break the traditional speed–fidelity tradeoff. We connect 13 papers published between 2018 and 2025, culminating in the heteronuclear Cs–Rb architecture of Farouk et al. (2023) and the analytical counter-diabatic pulses of Beterov et al. (2025).

Keywords. Rydberg atoms · CNOT gate · two-photon excitation · counter-diabatic driving · Förster resonance

1 The Problem

A quantum computer needs two ingredients: qubits that stay coherent, and gates that entangle them reliably. Neutral atoms — individual rubidium or cesium atoms held in focused laser beams called **optical tweezers** — excel at the first task. Their ground-state hyperfine levels provide long-lived qubit states ($|0\rangle$ and $|1\rangle$), and hundreds of atoms can be arranged in programmable arrays [1].

The hard part is the second task: making two atoms talk to each other on demand. Ground-state atoms barely interact at micron-scale separations. But if you laser-excite one atom to a **Rydberg state** — an orbital with principal quantum number $n \approx 50 - 100$, where the electron is hundreds of atomic radii from the nucleus — the atom becomes enormous and exquisitely sensitive to its neighbors. This conditional interaction is the basis of every Rydberg two-qubit gate.

The central question of this note is: **How do you design the laser pulses that turn Rydberg interactions into a high-fidelity CNOT gate?**

2 Background: Rydberg Blockade and the CZ Gate

2.1 The blockade mechanism

When two atoms are both in Rydberg states, they interact via a van der Waals potential $V(R) = C_6/R^6$, where R is their separation. If this interaction energy is much larger than the

laser Rabi frequency ($V \gg \Omega$), the doubly-excited state $|rr\rangle$ is shifted so far off resonance that the laser cannot excite both atoms simultaneously. This is the **Rydberg blockade** [2].

The blockade creates conditional dynamics: a laser pulse that would excite atom B to $|r\rangle$ is blocked if atom A is already in $|r\rangle$, but proceeds normally if A is in a ground state. This asymmetry is exactly what a controlled gate needs.

2.2 From blockade to CZ

The simplest Rydberg gate protocol works like this:

(i) Apply a π-pulse to atom A: $|1\rangle \to |r\rangle$.
(ii) Apply a 2π-pulse to atom B: $|1\rangle \to |r\rangle \to |1\rangle$, picking up a π phase.
(iii) Apply a $(-\pi)$-pulse to atom A: $|r\rangle \to |1\rangle$.

If A was in $|1\rangle$, step 2 is blocked and B gets no phase. If A was in $|0\rangle$, B completes its 2π rotation and picks up a -1 phase. The result is a **controlled-Z** (CZ) gate: $|11\rangle \to -|11\rangle$, all other basis states unchanged. A CNOT is then obtained by wrapping the CZ with Hadamard gates on the target qubit.

This three-step protocol has a problem: **atom A sits in the fragile Rydberg state during step 2**. Any decay or dephasing during that waiting time degrades fidelity. This motivates the search for better pulse designs.

2.3 CZ versus native CNOT

Most proposals construct CNOT indirectly: CZ + single-qubit rotations. But some schemes produce a **native CNOT** — a single pulse sequence whose truth table is directly that of CNOT without extra rotations. The EIT-based approach of the seed paper [3] and the proposal of Li et al. [4] are examples. Shi's transition slow-down protocol [5] is another. Native CNOT gates can be faster and simpler, but they often require more complex level structures.

3 Thread 1: Stark-Tuned Förster Resonances

3.1 Beyond van der Waals: Förster resonances

The C_6/R^6 van der Waals interaction arises when two Rydberg atoms are **not** at a Förster resonance — when the energy of state $|n_1S, n_2S\rangle$ differs from the nearest pair state $|n_{1'}P, n_{2'}P\rangle$ by a nonzero defect δ_0. But certain combinations of Rydberg states are nearly degenerate: $\delta_0 \approx 0$. At exact resonance, the interaction switches from $1/R^6$ to $1/R^3$ (dipole-dipole), which is **much stronger** at typical experimental distances of $5 - 30\mu m$.

3.2 Beterov 2018: The foundational protocol

Beterov et al. [6] proposed exploiting Förster resonances for two-qubit gates in cesium. The key idea: the Förster resonance $|90S_{1/2}, 96S_{1/2}\rangle \leftrightarrow |90P_{1/2}, 95P_{1/2}\rangle$ has an energy defect $\delta_0/2\pi = 75.6$ MHz in zero electric field. By applying a dc electric field of $E = 29.75$ mV/cm, the Stark effect tunes the defect to zero — hence **Stark-tuned Förster resonance**.

The gate protocol uses **double adiabatic rapid passage** (ARP): sweep the electric field slowly through resonance, let the system evolve and pick up a controlled phase, then sweep back. The adiabaticity ensures robustness — small variations in field strength or atom position don't ruin the gate.

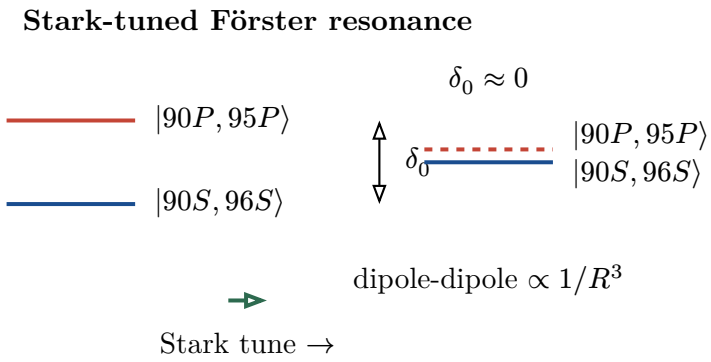

Figure 1: Stark-tuned Förster resonance. Left: the pair states $|90S, 96S\rangle$ and $|90P, 95P\rangle$ in Cs are separated by a Förster defect δ_0. A dc electric field Stark-shifts the levels to resonance (right), turning on strong $1/R^3$ dipole-dipole coupling [6].

3.2.1 Results and significance

The protocol achieves CNOT truth-table fidelity > 0.99 at interatomic distance $R = 25\mu m$, with gate time $\approx 1.8\mu s$. Crucially, it works at **large separations** — the $1/R^3$ Förster interaction doesn't need atoms to be as close as blockade-based gates ($\approx 5\mu m$). The fidelity is robust to $\pm 1\mu m$ position uncertainty.

The concept described here — **Stark-Tuned Resonance Adiabatic Passage** — is what we refer to as the STRAP approach throughout this note (see §9 for a caveat on this acronym).

3.3 Connection to the seed paper

Farouk et al. [3] extend this Förster-based approach to **heteronuclear** atom pairs (Cs control, Rb target), using the same group's expertise in Stark-tuned Förster resonances. The innovation is that heteronuclear Förster interactions allow parallel gate operations — the Cs control atom interacts with Rb targets but the Rb targets don't interact with each other (different species, different resonance conditions).

4 Thread 2: Two-Photon Excitation Pathways

4.1 Why two photons?

Excitation from a ground state ($|5S\rangle$ in Rb, $|6S\rangle$ in Cs) directly to a Rydberg $|nS\rangle$ state requires light in the deep ultraviolet. That's experimentally painful: UV lasers are noisy, and UV photons scatter off everything. The solution used by essentially all modern experiments is **two-photon excitation**: use two infrared/visible lasers through an intermediate P state.

$$|\text{ground } S\rangle \xrightarrow{\omega_1} |\text{intermediate } P\rangle \xrightarrow{\omega_2} |nS \text{ or } nD\rangle$$

If the intermediate state is far-detuned (detuning $\Delta \gg \Gamma_P$, where Γ_P is its decay rate), the atom never significantly populates it, and the effective dynamics look like a single transition with Rabi frequency $\Omega_{\text{eff}} = \Omega_1\Omega_2/(2\Delta)$. But the intermediate state still matters: its finite lifetime τ_P causes off-resonant scattering that limits gate fidelity.

4.2 The level structures across papers

Different groups use different atomic species and intermediate states. Table 1 compares them.

Table 1: Two-photon Rydberg excitation pathways used across the collected references. The seed paper [3] uses Cs for the control atom and Rb for the target — combining both level structures in a single gate.

Paper	Species	Ground → Intermediate	Intermediate → Rydberg	Lifetime
Saffman 2020 [2]	Cs	$6S_{1/2} \to 7P$	$7P \to nS_{1/2}$	155 ns
Pelegrí 2022 [7]	Cs	$6S_{1/2} \to 7P$	$7P \to nS/nD$	155 ns
Li 2022 [8]	^{87}Rb	$5S_{1/2} \to 6P_{3/2}$	$6P_{3/2} \to nS$	118 ns
Beterov 2025 [9]	Rb	$5S_{1/2} \to 6P_{3/2}$	$6P_{3/2} \to nS/nD$	118 ns
Beterov 2025 [9]	Cs	$6S_{1/2} \to 7P_{1/2}$	$7P_{1/2} \to nR$	155 ns
Farouk 2023 [3]	Cs (ctrl)	$6S_{1/2} \to 7P$	$7P \to 81S_{1/2}$	155 ns
Farouk 2023 [3]	Rb (tgt)	$5S_{1/2} \to 6P_{3/2}$	$6P_{3/2} \to 77S_{1/2}$	118 ns

The pattern is clear: **Cs goes through $7P$ (459 nm first photon), Rb goes through $6P_{3/2}$ (420 nm first photon)**. Both use a second infrared photon ($\approx 1010 - 1040$ nm) to reach high-n Rydberg states. The seed paper [3] is unique in using **both** species in the same gate.

4.3 Key results from two-photon gate proposals

Saffman et al. 2020 [2] designed symmetric CZ gates using adiabatic rapid passage (ARP) pulses with Cs two-photon excitation. The "symmetric" label means **both atoms receive identical pulses** — no individual addressing required. Using smooth super-Gaussian envelopes with sinusoidal detuning chirps, they achieved Bell state fidelity $F = 0.9994$ at gate time $T = 0.54\mu s$ with blockade $B/2\pi = 3$ GHz.

Pelegrí et al. 2022 [7] extended two-photon ARP to **multiqubit** gates (CCZ, three-qubit). They carefully modeled the full Cs hyperfine manifold, accounting for AC Stark shifts and spontaneous decay from the intermediate $7P$ state, achieving CCZ fidelity > 0.995 in $\approx 1.8\mu s$.

Li, Shao, Li 2022 [8] proposed a single-pulse CZ gate using the $6P$ state of ^{87}Rb — the same intermediate state as the seed paper's targets. The idea: a carefully shaped temporal pulse drives both the ground-to-Rydberg and Rydberg-to-ground transitions in one shot. Fidelity: $> 99.7\%$ in $< 1\mu s$.

Li, Qian, Zhang 2023 [4] proposed a **native CNOT** gate — not CZ-plus-Hadamards, but a single pulse sequence whose truth table is directly CNOT — using optimized smooth Gaussian-shaped pulses with two-photon Rydberg excitation. The key difference from CZ-based approaches: a native CNOT needs fewer total pulses and avoids the overhead of single-qubit correction gates. Their pulse optimization uses the two-photon pathway through an intermediate P state, analogous to the other proposals in Table 1.

Beterov et al. 2025 [9] is the first to analyze **three-photon** Rydberg excitation: $|5S\rangle \to |5P_{3/2}\rangle \to |7S_{1/2}\rangle \to |r\rangle$ in Rb, using standard D2 light (780 nm) as the first photon. This enables Doppler-free excitation geometry.

5 Thread 3: Breaking the Speed–Fidelity Tradeoff

5.1 The adiabatic dilemma

Adiabatic protocols — like the ARP pulses in Saffman 2020 [2] and the Förster sweeps in Beterov 2018 [6] — are **robust**. Small errors in laser intensity or atom position don't matter

much, because the system follows an eigenstate. But adiabaticity demands slowness: the pulse duration T must satisfy $T \gg 1/\Omega_{\max}$, or equivalently, the sweep rate must be much slower than the energy gap squared. Longer gate times mean more decoherence from Rydberg-state decay.

This creates a tension:

- **Slow and robust** (adiabatic): high intrinsic fidelity, but long exposure to decoherence.
- **Fast and fragile** (resonant π-pulses): short gate time, but sensitive to every experimental imperfection.

5.2 The time-optimal benchmark

Jandura and Pupillo [10] established the fundamental speed limit. Using semi-analytical optimal control with smooth pulse ansätze, they found the minimum gate time $T_{\min}$ set by the blockade strength B:

$$T_{\min} \approx \frac{4\pi}{B}$$

Their CZ gate uses global laser pulses (no single-site addressing), reaching the time-optimal bound. This provides a **benchmark**: any gate protocol can be compared against $T_{\min}$ to see how much overhead it adds.

A different route to fast gates is Shi's **transition slow-down** protocol [5]. Instead of optimizing the pulse shape to follow an adiabatic path faster, Shi exploits the blockade interaction itself: when the control atom is in $|r\rangle$, the target's transition slows down because the blockade shifts it off resonance. This differential evolution rate between the blocked and unblocked cases directly implements CNOT-like conditional dynamics — a native fast CNOT without the CZ detour.

5.3 Counter-diabatic driving: the best of both worlds

Counter-diabatic (CD) driving — also called **shortcuts to adiabaticity** or **transitionless quantum driving** — is a technique from quantum control theory. The idea is elegant: if you know the exact form of the diabatic errors (the transitions the system would make if you drove it too fast), you can add an extra field that **exactly cancels** those errors.

For a system with Hamiltonian $H_0(t)$ that you want to evolve adiabatically, the counter-diabatic Hamiltonian is:

$$H_{\mathrm{CD}(t)} = i\hbar \sum_n (|\partial_t n(t)\rangle\langle n(t)| - \langle n(t) \mid \partial_t n(t)\rangle \, |n(t)\rangle\langle n(t)|)$$

where $|n(t)\rangle$ are the instantaneous eigenstates. Driving with $H_0(t) + H_{\mathrm{CD}(t)}$ produces **exact adiabatic evolution at any speed**.

The catch: H_{CD} often requires control fields that are experimentally impossible (e.g., direct coupling between states with no allowed transition). The recent breakthroughs are in finding **approximate, experimentally realizable** versions.

5.3.1 Dalal and Sanders 2022

Dalal and Sanders [11] applied transitionless quantum driving to Rydberg CZ gates, using simultaneous broadband laser driving of the atom pair. This was an early proof-of-concept that CD techniques could be adapted to the Rydberg gate setting.

5.3.2 Yagüe Bosch et al. 2024

Yagüe Bosch et al. [12] made the key breakthrough: **effective counter-diabatic (eCD) driving** via Floquet engineering. They took the Saffman 2020 adiabatic protocol and accelerated it by a factor of $\approx 2\times$, reaching gate time $T = 0.26\mu s$ (down from $0.54\mu s$) with fidelity $F > 0.998$.

Their critical insight: rather than adding H_{CD} on top of the original H_0, they implemented **only** H_{CD} (approximated via oscillating fields). This avoids a subtle problem — the blockade-dependent dynamical phase $\varphi = \int \Omega^2(t)/(4V)\,\mathrm{d}t$ that accumulates during the adiabatic protocol and degrades fidelity when the blockade V is not infinite.

5.3.3 Beterov et al. 2025: Analytical CD pulses

Beterov et al. [9] (same group as the seed paper) achieved the most complete result: **fully analytical** counter-diabatic pulse shapes for symmetric CZ gates. The pulse parameters depend on a single number — the gate duration T:

$$\Omega_0^{\max} = \frac{4.006\pi}{T}, \quad \delta_0 = \frac{2\pi}{T}$$

with super-Gaussian envelope $\Omega_0(t) = \Omega_0^{\max} \exp\left[-(t-t_0)^4/w^4\right]$ and sinusoidal detuning $\delta(t) = \delta_0 \sin[2\pi(t-t_0)/T]$. The counter-diabatic correction adds a quadrature component:

$$\Omega_{\mathrm{CD}(t)} = -\frac{\dot{\Omega}_0\delta - \Omega_0\dot{\delta}}{\Omega_0^2 + \delta^2}$$

For single-photon excitation in Rb with $110P$ Rydberg states, this achieves $F = 0.9999$ in $T \approx 0.1\mu s$ — competitive with numerically optimized time-optimal protocols, but with **analytical** pulse shapes that are easier to implement and understand.

For two-photon excitation, each laser field becomes $\Omega_{1,2}(t) = \sqrt{2\Delta} \cdot \left[\Omega_0(t) - i\Omega_{\mathrm{CD}(t)}\right]$, where Δ is the intermediate-state detuning. This is directly applicable to the Cs and Rb level structures in Table 1.

6 Experimental Milestones

6.1 McDonnell et al. 2022: First EIT-based CNOT

McDonnell, Keary, and Pritchard [13] demonstrated the first **native CNOT** gate using electromagnetically induced transparency (EIT) with Cs atoms separated by $6\mu m$. The EIT protocol is the same one used in the seed paper [3]: the control atom's Rydberg excitation disrupts the EIT condition on the target atom, flipping its state.

Loss-corrected fidelity: $F_{\mathrm{CNOT}} \approx 0.82$. This is far below the fault-tolerance threshold, but it validated the EIT-based native CNOT concept experimentally.

6.2 Evered et al. 2023: State of the art

The Harvard/QuEra team [1] achieved the current record: 99.5% two-qubit CZ gate fidelity on up to 60 atom pairs in parallel. Key ingredients:

- **Optimal control**: numerically optimized single-pulse gate shapes.
- **Atomic dark states**: laser polarization chosen to minimize off-resonant scattering from intermediate states.
- **Parallel operation**: global laser beams, no single-site addressing.

This result surpasses the surface-code error-correction threshold, demonstrating that Rydberg gates can in principle support fault-tolerant quantum computing.

7 The Seed Paper: Convergence

Farouk et al. [3] synthesize multiple threads into a single architecture for parallel multi-qubit CNOT gates.

7.1 Architecture

- **Control atom (Cs):** ground states $|6S_{1/2}, F = 3\rangle$, $|6S_{1/2}, F = 4\rangle \rightarrow$ Rydberg $|81S_{1/2}\rangle$ via two-photon excitation (460 nm: $6S \rightarrow 7P$, 1013 nm: $7P \rightarrow 81S$).
- **Target atoms (Rb):** ground states $|5S_{1/2}, F = 1\rangle$, $|5S_{1/2}, F = 2\rangle \rightarrow$ intermediate $|6P_{3/2}\rangle \rightarrow$ Rydberg $|77S_{1/2}\rangle$ (420 nm + 1039 nm). Inverted-Y EIT configuration.

7.2 Gate mechanism

The CNOT operates via EIT and Rydberg blockade. When the Cs control is in $|1\rangle$ (not excited to Rydberg), the Rb target experiences a transparent EIT window and its state is unchanged. When the Cs control is in $|0\rangle$ (mapped to Rydberg), the blockade breaks EIT on the Rb target, causing a state flip.

The innovation is **heteronuclear Förster resonance**: the Cs–Rb interaction (control–target) uses Förster coupling at $1/R^3$, while the Rb–Rb interaction (target–target) uses the weaker van der Waals at $1/R^6$. This separation of interaction scales enables:

- $\mathbf{CNOT^N}$: one Cs control atom flips N Rb targets in parallel.
- $\boldsymbol{C_2}\mathbf{NOT^2}$: two Cs controls jointly gate two Rb targets (Toffoli-like).

7.3 What it draws from

The seed paper sits at the intersection of Threads 1 and 2, plus an experimental proof-of-concept:

- **Thread 1 (Förster):** Heteronuclear Stark-tuned Förster resonances from Beterov's group [6].
- **Thread 2 (Two-photon):** Both Cs and Rb use two-photon Rydberg excitation through intermediate P states (Table 1).
- **Experimental validation:** The EIT-based native CNOT protocol demonstrated by McDonnell et al. [13].

The seed paper does **not** yet use counter-diabatic techniques (Thread 3). That combination — heteronuclear Förster gates with CD pulse engineering — is an open frontier (Section 5.3, §9).

8 Timeline and Field Evolution

Figure 2 traces the chronological development.

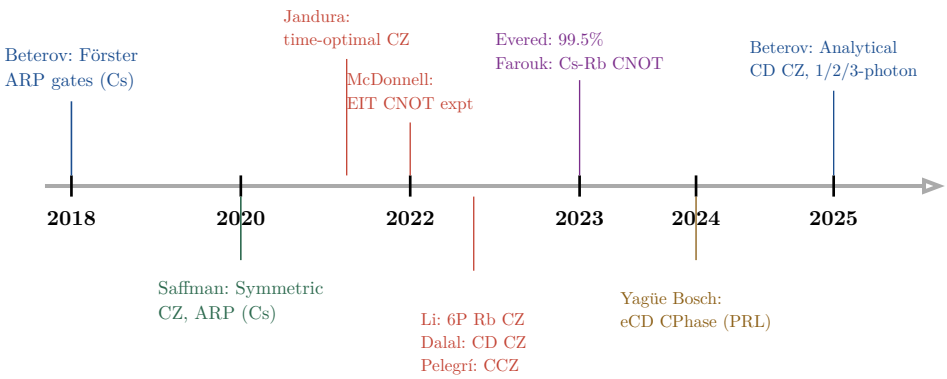

Figure 2: Timeline of Rydberg CNOT/CZ gate development. The field progresses from slow adiabatic Förster protocols (2018) through symmetric adiabatic gates and time-optimal benchmarks (2020–2022), to counter-diabatic shortcuts (2024–2025), with experimental milestones in 2022–2023.

Table 2: Comparison of Rydberg gate proposals. Fidelities for proposals are theoretical (no experimental imperfections); experimental results from Evered and McDonnell include real errors.

Paper	Gate	Species	Protocol	Best Fidelity	Gate Time
Beterov 2018 [6]	CZ/CNOT	Cs	Double ARP, Förster	> 0.99	$1.8\mu s$
Saffman 2020 [2]	CZ	Cs	Symmetric ARP	0.9994	$0.54\mu s$
Jandura 2022 [10]	CZ	generic	Time-optimal	fund. limit	$T_{\min} = 4\pi/B$
Yagüe Bosch 2024 [12]	CZ	generic	eCD shortcut	> 0.998	$0.26\mu s$
Beterov 2025 [9]	CZ	Rb/Cs	Analytical CD	0.9999	$\approx 0.1\mu s$
Li 2023 [4]	CNOT	generic	Two-photon optimized	high	theory
Shi 2022 [5]	CNOT	generic	Transition slow-down	high	theory
Evered 2023 [1]	CZ	^{87}Rb	Optimal control	0.995	expt.
McDonnell 2022 [13]	CNOT	Cs	EIT blockade	0.82	expt.

9 Open Questions and Gaps

Several questions remain open in this landscape:

(i) **STRAP as a term.** Our literature search found no paper using "STRAP" as a standard acronym. The concept it likely refers to — Stark-Tuned Resonance Adiabatic Passage — is well-established in Beterov et al. [6], but the abbreviation may be informal or used in a specific group's internal nomenclature. Clarification is needed.

(ii) **Counter-diabatic CNOT.** All counter-diabatic proposals to date target **CZ** gates [9,11,12]. No counter-diabatic protocol has been proposed (or demonstrated) specifically for a native CNOT. Since CNOT = Hadamard · CZ · Hadamard, this is not a fundamental barrier, but a native CD-CNOT might offer advantages.

(iii) **Experimental demonstration of CD gates.** Counter-diabatic Rydberg gates remain theoretical proposals. The experimental state of the art [1] uses numerically optimized pulses, not CD-derived pulse shapes. Demonstrating CD advantages (robustness + speed) in a real experiment is the next milestone.

(iv) **Heteronuclear CD protocols.** The seed paper [3] uses heteronuclear Cs–Rb interactions. The CD protocols [9,12] assume homonuclear interactions (identical atoms). Extending counter-diabatic driving to heteronuclear Förster gates is unexplored.

(v) **Scaling to many qubits.** Pelegrí et al. [7] extended two-photon ARP to three-qubit CCZ gates. The seed paper [3] demonstrated parallel CNOT^N architectures. How counter-diabatic techniques perform for multi-qubit gates beyond two remains open.

(vi) **Three-photon excitation.** Beterov et al. [9] introduced three-photon Rydberg excitation ($5S \to 5P \to 7S \to nR$ in Rb), which enables Doppler-free geometry. No experiment has demonstrated three-photon Rydberg gates.

10 Summary

The story of Rydberg CNOT gate design is one of convergence:

- **Stark-tuned Förster resonances** [6] provided the foundational interaction mechanism — strong $1/R^3$ coupling, controllable via external electric fields, robust to position uncertainty.
- **Two-photon excitation** made Rydberg states experimentally accessible. The Cs ($6S \to 7P \to nS$) and Rb ($5S \to 6P \to nS$) pathways are now standard, and the seed paper [3] uniquely combines both in a heteronuclear architecture.
- **Counter-diabatic driving** [9,12] resolves the speed–fidelity tradeoff that limited adiabatic protocols. Analytical pulse shapes now achieve $F = 0.9999$ at gate times competitive with numerically optimized time-optimal protocols [10].
- **Experiments** validated the EIT-based CNOT concept [13] and proved that Rydberg gates can surpass the fault-tolerance threshold [1].

The frontier is clear: combine heteronuclear architectures, counter-diabatic pulse design, and the latest experimental techniques into a single platform. The seed paper [3] points toward this synthesis; the counter-diabatic tools from Beterov et al. [9] provide the pulse-engineering machinery; and the Harvard/QuEra results [1] show the experimental path is open.

Bibliography

[1] S. J. Evered et al., High-fidelity parallel entangling gates on a neutral-atom quantum computer, Nature **622**, 268 (2023).

[2] M. Saffman, I. I. Beterov, A. Dalal, E. J. Páez, and B. C. Sanders, Symmetric Rydberg controlled-Z gates with adiabatic pulses, Physical Review a **101**, 62309 (2020).

[3] A. M. Farouk, I. I. Beterov, P. Xu, S. Bergamini, and I. I. Ryabtsev, Parallel Implementation of CNOT^N and C_2NOT^2 Gates via Homonuclear and Heteronuclear Förster Interactions of Rydberg Atoms, Photonics **10**, 1280 (2023).

[4] R. Li, J. Qian, and W. Zhang, Proposal for practical Rydberg quantum gates using a native two-photon excitation, Arxiv Preprint Arxiv:2303.16605 (2023).

[5] X.-F. Shi, Transition Slow-Down by Rydberg Interaction of Neutral Atoms and a Fast Controlled-NOT Quantum Gate, Physical Review Applied **18**, 34072 (2022).

[6] I. I. Beterov, M. Saffman, E. A. Yakshina, D. B. Tretyakov, V. M. Entin, S. Bergamini, E. A. Kuznetsova, and I. I. Ryabtsev, Two-qubit gates using adiabatic passage of the Stark-tuned Förster resonances in Rydberg atoms, Physical Review a **97**, 32701 (2018).

[7] G. Pelegr\ĭ, A. J. Daley, and J. D. Pritchard, High-fidelity multiqubit Rydberg gates via two-photon adiabatic rapid passage, Physical Review a **106**, 42616 (2022).

[8] X. X. Li, X. Q. Shao, and W. Li, Single temporal-pulse-modulated parameterized controlled-phase gate for Rydberg atoms, Physical Review Applied **18**, 44042 (2022).

[9] I. I. Beterov, K. V. Kozenko, P. Xu, and I. I. Ryabtsev, Symmetric C_Z gate for ultracold neutral atoms based on counterdiabatic driving at Rydberg excitation, Arxiv Preprint Arxiv:2510.04766 (2025).

[10] S. Jandura and G. Pupillo, Time-Optimal Two- and Three-Qubit Gates for Rydberg Atoms, Quantum **6**, 712 (2022).

[11] A. Dalal and B. C. Sanders, Two-qubit gate in neutral atoms using transitionless quantum driving, Arxiv Preprint Arxiv:2206.08915 (2022).

[12] L. S. Yagüe Bosch, T. Ehret, F. Petiziol, E. Arimondo, and S. Wimberger, Shortcut-to-Adiabatic Controlled-Phase Gate in Rydberg Atoms, Physical Review Letters **132**, 100602 (2024).

[13] K. McDonnell, L. F. Keary, and J. D. Pritchard, Demonstration of a Quantum Gate using Electromagnetically Induced Transparency, Physical Review Letters **129**, 200501 (2022).